

PathoFMPred complete three-representation output

COAD TCGA-A6-A56B | TITAN, Giga-SSL and Prov-GigaPath | 2026-09-20

UNVALIDATED INTERNAL TCGA RESEARCH OUTPUT. BINARY REFERENCE RANK IS NOT PROBABILITY. DO NOT USE FOR DIAGNOSIS, TREATMENT SELECTION OR CLINICAL RISK ESTIMATION.

Output inventory

Representation	Continuous outputs	Binary outputs	Slides pooled	Output rule
TITAN	15	8	1	Q2 >= 0.20; AUROC >= 0.60
Giga-SSL	12	7	1	Q2 >= 0.20; AUROC >= 0.60
Prov-GigaPath	14	7	1	Q2 >= 0.20; AUROC >= 0.60

Case context, not used as model input

The sigmoid-colon slide contained a moderately differentiated adenocarcinoma extending through the muscularis propria without reaching the serosal surface (pT3). The excision margins were negative, and carcinoma was present in three of 25 regional lymph nodes.

Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3).

How to read the report

- PathoFMPred lists only representation-specific fitted objects that crossed the recorded internal performance threshold and passed the package's default sample-size rule.
- Continuous predictions remain in the source endpoint units. The internal TCGA rank provides display context and is not a clinical reference interval.
- Binary calls use an uncalibrated fitted LDA rule. The original LDA score is model-specific; neither the score nor its TCGA rank is a probability.
- The example patients lack reference labels for the displayed outcomes in the processed analysis table, so this report demonstrates software output rather than correctness.

Feature group	Reference label source
Mutations	TCGA MC3 protein-altering calls within the Bailey tissue-specific driver catalogue
Genome doubling and aneuploidy	Taylor et al. arm-level and genome-doubling variables
MSI	cBioPortal TCGA PanCancer MANTIS and MSIsensor variables
Oncogenic pathways	Sanchez-Vega pathway alteration matrix
Immune and genomic context	Published Thorsson PanImmune computational phenotypes

TITAN: complete predictable output

This page reports all 15 continuous and 8 binary outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score	14.857372	source endpoint units	68.3	0.302	0.575	408	Retained under code grouping
MSI	MANTIS score	0.341518	source endpoint units	26.7	0.284	0.4	391	Code-group sensitive: below original documented screening threshold
MSI	MSIsensor score	-2.105765	source endpoint units	10.8	0.222	0.253	411	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_ALLOGRAFT_REJECTION	-0.428158	source endpoint units	4.4	0.233	0.474	424	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_IL6_JAK_STAT3_SIGNALING	-0.428552	source endpoint units	2.5	0.206	0.47	424	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_INTERFERON_ALPHA_RESPONSE	-0.553623	source endpoint units	2.9	0.215	0.475	424	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_INTERFERON_GAMMA_RESPONSE	-0.526827	source endpoint units	2.2	0.251	0.507	424	Code-group sensitive: below original documented screening threshold
Rna Pathway Activity	HALLMARK_MYOGENESIS	-0.314861	source endpoint units	7.2	0.22	0.441	424	Code-group sensitive: below original documented screening threshold
Immune/context	Aneuploidy Score	13.162893	source endpoint units	74.9	0.217	0.518	408	Retained under code grouping
Immune/context	Lymphocyte Infiltration Signature Score	-0.87866	source endpoint units	6.1	0.297	0.532	417	Retained under code grouping
Immune/context	Macrophage Regulation	-1.022325	source endpoint units	2.5	0.226	0.471	417	Code-group sensitive: near chance or worse
Immune/context	Nonsilent Mutation Rate	1.146518	log1p-transformed source endpoint units	17.2	0.275	0.345	345	Code-group sensitive: near chance or worse
Immune/context	SNV Neoantigens	3.688697	log1p-transformed source endpoint units	11.6	0.241	0.363	350	Code-group sensitive: below original documented screening threshold
Immune/context	Silent Mutation Rate	0.545944	log1p-transformed source endpoint units	17	0.259	0.368	345	Code-group sensitive: near chance or worse
Immune/context	TIL Regional Fraction	8.19411	source endpoint units	44.3	0.407	0.69	414	Retained under code grouping

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC	PR-AUC	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Absent	-0.231	58.5	0.735	0.644	408 (165/243)	Retained under code grouping
Mutation	APC	Mutation	2.111	76.6	0.851	0.822	382 (196/186)	Code-group sensitive: near chance or worse
Mutation	KRAS	Wild type	-0.988	56.9	0.759	0.559	382 (126/256)	Retained under code grouping
Mutation	TP53	Mutation	1.158	70.5	0.852	0.789	382 (161/221)	Retained under code grouping
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-4.947	44.1	0.824	0.626	391 (74/317)	Retained under code grouping
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-6.762	59	0.902	0.714	374 (57/317)	Retained under code grouping
Pathway	MYC	Not altered	-4.726	2.3	0.647	0.375	323 (79/244)	Code-group sensitive: near chance or worse
Pathway	TP53	Altered	1.985	76.3	0.734	0.815	323 (208/115)	Retained under code grouping

AUROC and PR-AUC are internal repeated out-of-fold performance estimates. The class call shown here comes from the fitted model object and is not a calibrated probability. Grouped sensitivity uses TCGA tissue-source-site code, a barcode-derived cohort variable rather than an institution, scanner or laboratory identifier.

Giga-SSL: complete predictable output

This page reports all 12 continuous and 7 binary outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score	18.221593	source endpoint units	90.3	0.26	0.514	408	Grouped sensitivity not evaluated
Aneuploidy	Deleted arm count	10.24538	source endpoint units	87.6	0.209	0.528	408	Grouped sensitivity not evaluated
MSI	MANTIS score	0.291088	source endpoint units	13.7	0.266	0.44	391	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_ALLOGRAFT_REJECTION	-0.351667	source endpoint units	12.2	0.312	0.581	424	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_IL6_JAK_STAT3_SIGNALING	-0.253947	source endpoint units	17.4	0.245	0.515	424	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_INTERFERON_GAMMA_RESPONSE	-0.450935	source endpoint units	6	0.277	0.543	424	Grouped sensitivity not evaluated
Immune/context	Aneuploidy Score	15.359554	source endpoint units	90.8	0.238	0.503	408	Grouped sensitivity not evaluated
Immune/context	Leukocyte Fraction	0.079558	source endpoint units	5.1	0.288	0.557	419	Grouped sensitivity not evaluated
Immune/context	Lymphocyte Infiltration Signature Score	-0.336206	source endpoint units	30.6	0.273	0.516	417	Grouped sensitivity not evaluated
Immune/context	Macrophage Regulation	-0.83572	source endpoint units	11.7	0.284	0.548	417	Grouped sensitivity not evaluated
Immune/context	SNV Neoantigens	3.596647	log1p-transformed source endpoint units	9.6	0.226	0.361	350	Grouped sensitivity not evaluated
Immune/context	TIL Regional Fraction	19.117364	source endpoint units	93.6	0.377	0.628	414	Grouped sensitivity not evaluated

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC	PR-AUC	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Present	1.538	89.3	0.763	0.676	408 (165/243)	Grouped sensitivity not evaluated
Mutation	APC	Mutation	1.845	62.4	0.795	0.73	382 (196/186)	Grouped sensitivity not evaluated
Mutation	KRAS	Mutation	-2.184	24.2	0.693	0.457	382 (126/256)	Grouped sensitivity not evaluated
Mutation	TP53	Wild type	-0.872	43.6	0.786	0.673	382 (161/221)	Grouped sensitivity not evaluated
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-3.795	65.5	0.777	0.533	391 (74/317)	Grouped sensitivity not evaluated
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-7.64	39.3	0.851	0.584	374 (57/317)	Grouped sensitivity not evaluated
Pathway	TP53	Altered	2.481	81.9	0.749	0.842	323 (208/115)	Grouped sensitivity not evaluated

AUROC and PR-AUC are internal repeated out-of-fold performance estimates. The class call shown here comes from the fitted model object and is not a calibrated probability. Grouped sensitivity uses TCGA tissue-source-site code, a barcode-derived cohort variable rather than an institution, scanner or laboratory identifier.

Prov-GigaPath: complete predictable output

This page reports all 14 continuous and 7 binary outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score	17.678433	source endpoint units	85.6	0.294	0.561	407	Grouped sensitivity not evaluated
Aneuploidy	Deleted arm count	12.369035	source endpoint units	96.7	0.205	0.539	407	Grouped sensitivity not evaluated
MSI	MANTIS score	0.273867	source endpoint units	12.4	0.3	0.458	390	Grouped sensitivity not evaluated
MSI	MSIsensor score	-2.185303	source endpoint units	11.1	0.254	0.327	410	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_ALLOGRAFT_REJECTION	-0.594614	source endpoint units	3.4	0.318	0.581	423	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_IL6_JAK_STAT3_SIGNALING	-0.489376	source endpoint units	4.4	0.244	0.533	423	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_INTERFERON_ALPHA_RESPONSE	-0.762535	source endpoint units	0.1	0.231	0.486	423	Grouped sensitivity not evaluated
Rna Pathway Activity	HALLMARK_INTERFERON_GAMMA_RESPONSE	-0.682097	source endpoint units	1.1	0.284	0.566	423	Grouped sensitivity not evaluated
Immune/context	Aneuploidy Score	14.624235	source endpoint units	85.4	0.296	0.562	407	Grouped sensitivity not evaluated
Immune/context	Leukocyte Fraction	0.075621	source endpoint units	6.1	0.274	0.538	418	Grouped sensitivity not evaluated
Immune/context	Lymphocyte Infiltration Signature Score	-0.902111	source endpoint units	8.8	0.311	0.565	416	Grouped sensitivity not evaluated
Immune/context	Macrophage Regulation	-0.853166	source endpoint units	16.9	0.291	0.553	416	Grouped sensitivity not evaluated
Immune/context	SNV Neoantigens	3.449005	log1p-transformed source endpoint units	4.7	0.271	0.356	350	Grouped sensitivity not evaluated
Immune/context	TIL Regional Fraction	8.818737	source endpoint units	55.1	0.507	0.689	413	Grouped sensitivity not evaluated

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC	PR-AUC	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Present	2.12	88.1	0.765	0.701	407 (165/242)	Grouped sensitivity not evaluated
Mutation	APC	Mutation	4.716	94.6	0.842	0.796	382 (196/186)	Grouped sensitivity not evaluated
Mutation	KRAS	Mutation	-0.355	62.2	0.74	0.526	382 (126/256)	Grouped sensitivity not evaluated
Mutation	TP53	Mutation	3.277	96.7	0.787	0.695	382 (161/221)	Grouped sensitivity not evaluated
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-5.655	27.8	0.822	0.618	390 (73/317)	Grouped sensitivity not evaluated
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-9.736	22.4	0.857	0.62	373 (56/317)	Grouped sensitivity not evaluated
Pathway	TP53	Altered	4.208	89.3	0.724	0.809	323 (208/115)	Grouped sensitivity not evaluated

AUROC and PR-AUC are internal repeated out-of-fold performance estimates. The class call shown here comes from the fitted model object and is not a calibrated probability. Grouped sensitivity uses TCGA tissue-source-site code, a barcode-derived cohort variable rather than an institution, scanner or laboratory identifier.